MINOR VARIATION NOTIFICATION FORM FOR FULL EVALUATION PRODUCTS

Please read the following instructions before filling in the form.

1) This notification form is for Product Registration Holder (PRH) to notify National Pharmaceutical Control Bureau (NPCB) on the implementation of MiV-Notification (MiV-N) as per Malaysian Variation Guideline (MVG). The timeline for NPCB to acknowledge the variation notification is within 20 working days following receipt of a notification.
2) Please refer to the MVG for the conditions and supporting documents required.
3) Submission of relevant revised draft of package insert and labeling is subject to current regulatory requirements as per the latest Drug Registration Guidance Document (DRGD) and Circulars from NPCB. In the event that the revised draft of package insert and labeling does not meet the current regulatory requirements, please submit under Minor Variation Prior Approval (MiV-PA) or Major Variation (MaV).
4) PRH **must** submit this notification form together with the online submission through the Quest 2 system as the approval will only be notified via online submission. For submission online, please scan this form and attach together with the revised draft of package insert and labeling as a single file.
5) A MiV-N application **may be rejected** in specific circumstances with the consequence that the PRH must cease to apply the already implemented variation.
6) The completed form must be submitted to :
Seksyen Variasi, Pusat Pasca Pendaftaran Produk, Biro Pengawalan Farmaseutikal Kebangsaan, Kementerian Kesihatan Malaysia, Lot 36, Jalan Universiti, 46200 Petaling Jaya, Selangor. (Fax No: 03-79567151)

Product Name: **LOFRAL 10MG TABLET**	**Name and address of product registration holder:** **MEPHARM (MALAYSIA) SDN.BHD.** **15, JALAN USJ 1/19, TMN SUBANG PERMAI** **47600 SUBANG JAYA, SELANGOR**
Reference Number: 2006053027325	**Tel. No.: 03-5636 8088**
Registration Number: MAL07031138AC	**Fax No.: 03-5636 8133**
Date of online submission in Quest System: 4/12/13	**Email address: simonwong@mepharm.com**

Variation No.	**Minor Variation (Notification)**	**Please tick (√)** *Multiple selection is allowed*
MiV-N1	Change of details of product registration holder	
MiV-N2	Change of importer and/or store address	
MiV-N3	Change of product owner	✓
MiV-N4	Change in ownership of manufacturer	
MiV-N5	Change of the name or address (for example: postal code, street name) of the manufacturer of drug product	
MiV-N6	Change of the name or address (for example: postal code, street name) of the company or manufacturer responsible for batch release	
MiV-N7	Change of the name and/or address (for example: postal code, street name) of a manufacturer of the drug substance	
MiV-N8	Withdrawal/deletion of the alternative manufacturer(s) for drug substance	
MiV-N9	Renewal of European Pharmacopoeial Certificate of Suitability (CEP)	
MiV-N10	Change of specifications of the drug product and/or drug substance and/or excipient, following the updates in the compendium	
MiV-N11	Deletion of pack size for a product	

I hereby notify NPCB on the minor variation by notification for the product(s) above and declare that

- ✓ There is no other change except for the proposed variation;
- ✓ The change(s) will not adversely affect the quality, efficacy and safety of the product;
- ✓ All conditions for the variation concerned are fulfilled;
- ✓ The required supporting documents as specified for the variation in MVG have been submitted; and
- ✓ The proposed change has been checked in reference with the currently approved data in the system & there are no discrepancies.

WONG YOON YOU

Mepharm (Malaysia) Sdn Bhd (349607-D)
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47600 Subang Jaya Selangor Darul Ehsan
Tel: (03) 5636 8088 Fax: (603) 5636 8133

Name	Signature	Date
WONG YOON YOU		4/12/13

LOFRAL™

LOFRAL™ -5 tablets with amlodipine 5mg
LOFRAL™ -10 tablets with amlodipine 10mg

Qualitative and quantitative composition

LOFRAL™ -5 tablets
Each tablet contains:
Amlodipine besylate ……………………………........6.944mg
(↔ 5mg amlodipine base)
Microcrystalline cellulose…………………………124.056mg
Anhydrous dibasic calcium phosphate……………63.000mg
Sodium starch glycolate……………………………...4.000mg
Magnesium stearate…………………………………..2.000mg

LOFRAL™ -10 tablets
Each tablet contains:
Amlodipine besylate ……………………………........13.886mg
(↔ 10mg amlodipine base)
Microcrystalline cellulose……………………………248.111mg
Anhydrous dibasic calcium phosphate……………126.000mg
Sodium starch glycolate……………………………...8.000mg
Magnesium stearate…………………………………..4.000mg

Properties, effects

Pharmacology: Amlodipine is a calcium ion influx inhibitor (slow channel-blocker or calcium ion antagonist) and inhibits the transmembrane influx of calcium ions into the cardiac and vascular smooth muscle.
The mechanism of the antihypertensive action of amlodipine is due to a direct relaxant effect on the vascular smooth muscle. The precise mechanism by which amlodipine relieves angina has not been fully determined, but amlodipine reduces the total ischemic burden by the following 2 actions:
1) Amlodipine dilates the peripheral arterioles and thus, reduces the total peripheral resistance (afterload) against which the heart works. Since the heart rate remains stable, this unloading of the heart reduces myocardial energy consumption and oxygen requirements.
2) The mechanism of action of amlodipine probably involves dilation of the main coronary arteries and coronary arterioles, both in normal and ischemic regions. This dilatation increases myocardial oxygen delivery in patients with coronary artery spasm (Prinzmetal's or variant angina) and blunt smoking-induced coronary vasoconstriction.
In patients with hypertension, once-daily dosing provides clinically significant reductions of blood pressure in both the supine and standing positions throughout the 24-hr interval. Due to the slow onset of action, acute hypotension is not a feature of amlodipine administration.
In patients with angina, once-daily administration of amlodipine increases total exercise time, time to angina onset, and time to 1 mm ST segment depression and decreases both angina attack frequency and nitroglycerine tablet consumption.
Amlodipine has not been associated with any adverse metabolic effects or changes in plasma lipids and is suitable for use in patients with asthma, diabetes and gout.
Pharmacokinetics studies with cyclosporin have demonstrated that amlodipine does not significantly alter the pharmacokinetics of cyclosporin.
Hemodynamic studies and a controlled clinical trial in NYHA Class II-IV heart failure patients have shown that amlodipine did not lead to clinical deterioration as measured by exercise tolerance, left ventricular ejection fraction and clinical symptomatology.

Pharmacokinetics: After oral administration of therapeutic doses, amlodipine is well absorbed with peak blood levels between 6-12 hrs post-dose. Absolute bioavailability has been estimated to be between 64 and 80%. The volume of distribution was approximately 21 L/kg. Absorption of amlodipine is unaffected by consumption of food.
The terminal plasma elimination half-life is about 35-50 hrs and is consistent with once-daily dosing. Steady-state plasma levels are reached after 7-8 days of consecutive dosing. Amlodipine is extensively metabolized to inactive metabolites with 10% of the parent compound and 60% of metabolites excreted in the urine.
In vitro studies have shown that approximately 97.5% of circulating amlodipine is bound to plasma proteins.

Indications

First-line treatment of hypertension and as the sole agent to control blood pressure in the majority of patients. Patients not adequately controlled on a single antihypertensive agent may benefit from the addition of amlodipine, which has been used successfully in combination with a thiazide diuretic, β-adrenoceptor-blocking agent or an angiotensin-converting enzyme inhibitor.
Amlodipine is indicated for the 1st-line treatment of myocardial ischemia, whether due to fixed obstruction (stable angina) and/or vasospasm/vasoconstriction (Prinzmetal's or variant angina) of coronary vasculature. Amlodipine may be used where the clinical presentation suggests a possible vasospastic/vasoconstrictive component but where vasospasm/vasoconstriction has not been confirmed. Amlodipine may be used alone, as monotherapy or in combination with other antianginal drugs in patients with angina that is refractory to nitrates and/or adequate doses of β-blockers.

Recommended dosage

For both hypertension and angina, the usual initial dose is 5 mg once daily which may be increased to a maximum dose of 10 mg depending on the individual patient's response.
No dose adjustment of amlodipine is required upon concomitant administration of thiazide diuretics, β-blockers and angiotensin-converting enzyme inhibitors.
Elderly and in Renal Failure: Refer to Precautions.

Mode of Administration

Tablets for oral administration.
Take the tablet out of blister and swallow it with water.

Contraindications

Patients with known sensitivity to amlodipine or dihydropyridines.

Warnings and Precautions

Renal Failure: Amlodipine is extensively metabolized to inactive metabolites with 10% excreted as unchanged drug in the urine. Changes in amlodipine plasma concentrations are not correlated with degree of renal impairment. Amlodipine may be used in such patients at normal doses. Amlodipine is not dialysable.
Impaired Hepatic Function: As with all calcium antagonists, amlodipine half-life is prolonged in patients with impaired liver function and dosage recommendations have not been established. The drug should therefore be administered with caution in these patients.
Effects on the Ability to Drive or Operate Machinery: Not applicable.
Use in children: No clinical experience available on use of amlodipine in children.
Use in the elderly: The time to reach peak plasma concentrations of amlodipine is similar in elderly and younger subjects. Amlodipine clearance tends to be decreased with resulting increases in AUC and elimination half-life in elderly. Amlodipine, used at similar doses in elderly or younger patients, is equally well tolerated. Therefore, normal dosage regimens are recommended.

Pregnancy and lactation

Safety of amlodipine in human pregnancy or lactation has not been established. Amlodipine does not demonstrate toxicity in animal reproductive studies other than to delay parturition and prolong labor in rats at a dose level 50 times the maximum recommended dose in humans. Accordingly, use in pregnancy is only recommended when there is no safer alternative and when the disease itself carries greater risk for the mother and child.

Side effects

Amlodipine is well tolerated. In placebo-controlled clinical trials involving patients with hypertension or angina, the most commonly observed side effects were headache, edema, fatigue, nausea, flushing and dizziness. Less commonly observed side effects in marketing experience include pruritus, rash, dyspnea, asthenia, muscle cramps, dyspepsia, gingival hyperplasia and rarely, erythema multiforme. As with other calcium-channel blockers, the following adverse events have been rarely reported and cannot be distinguished from the natural history of the underlying disease, e.g. myocardial infarction, arrhythmia (including ventricular tachycardia and atrial fibrillation) and chest pain.
No pattern of clinically significant laboratory test abnormalities related to amlodipine has been observed.

Interactions

Amlodipine has been safely administered with thiazide diuretics, β-blockers, angiotensin-converting enzyme inhibitors, long-acting nitrates, sublingual nitroglycerin, NSAIDs, antibiotics and oral hypoglycemic drugs.
In healthy male volunteers, the co-administration of amlodipine does not significantly alter the effect of warfarin on prothrombin response time.
Studies have indicated that the co-administration of amlodipine with digoxin did not change serum digoxin levels or digoxin renal clearance in normal volunteers, and that co-administration of cimetidine did not alter the pharmacokinetics of amlodipine.
In vitro data from studies with human plasma indicate that amlodipine has no effect on protein-binding of the drugs tested (digoxin, phenytoin, warfarin or indomethacin).
Incompatibilities: Not relevant.

Overdosage

In humans, experience with intentional overdose is limited. Gastric lavage may be worthwhile in some cases. Available data suggest that gross overdosage could result in excessive peripheral vasodilation with subsequent marked and probably prolonged systemic hypotension. Clinically significant hypotension due to amlodipine overdosage calls for active cardiovascular support including monitoring of cardiac and respiratory function, elevation of extremities, and attention to circulating fluid volume and urine output. A vasoconstrictor may be helpful in restoring vascular tone and blood pressure, provided that there is no contraindication to its use. IV calcium gluconate may be beneficial in reversing the effects of calcium-channel blockade. Since amlodipine is highly protein-bound, dialysis is not likely to be of benefit.

Presentations

LOFRAL™ -5 tablets: White to off-white round tablets, with “A/5” embossed on break line side and “mp” on convex side.
Packing: The tablets are packed into blister pack of 10x3 per box.
LOFRAL™ -10 tablets: White to off-white round tablets, with “A/10” embossed on break line side and “mp” on convex side.
Packing: The tablets are packed into blister pack of 10x3 per box.

Other information

Storage: Do not store above 30°C.
Expiry: The preparation should not be used after the date marked “EXP” on the pack.
Shelf life: 2 years
Keep out of reach of children.

The information contained here is limited. Further information can be obtained from your doctor or pharmacist.

Information date
Nov 2013

Manufactured by:
Atlantic Pharma, S.A.
Zona Industrial da Abrunheira – Rua da Tapada Grande, 2
2710-089 Sintra, Portugal
for Acino, Liesberg, Basel-Land, Switzerland

acino
Switzerland